

Lab 2: Agile Backlog Creation & Sprint Simulation in Linear

Problem Statement #: Household Carbon Footprint & Target Tracker

Linear Project: HOUS (Household Carbon Footprint & Target Tracker)

Submission: Backlog, Epics, Sprints & Burndown Charts

Project Overview

This submission documents the Agile backlog and sprint simulation completed in Linear for the Household Carbon Footprint & Target Tracker project. The functional requirements were broken down into one Epic and three User Stories, prioritized, estimated using the Fibonacci scale, and executed across one simulated sprint.

Epic: Household Carbon Emission Tracking (HOUS-1)

Description: Enable users to record monthly household energy and fuel consumption, calculate the resulting CO2e emissions, and review historical emission data over time.

Backlog & User Stories

The backlog contains three User Stories under the Household Carbon Emission Tracking epic: Record Monthly Consumption (3 pts), Calculate Monthly CO2e Emissions (5 pts), and View Historical Emissions (3 pts), for a total estimate of 11 story points.

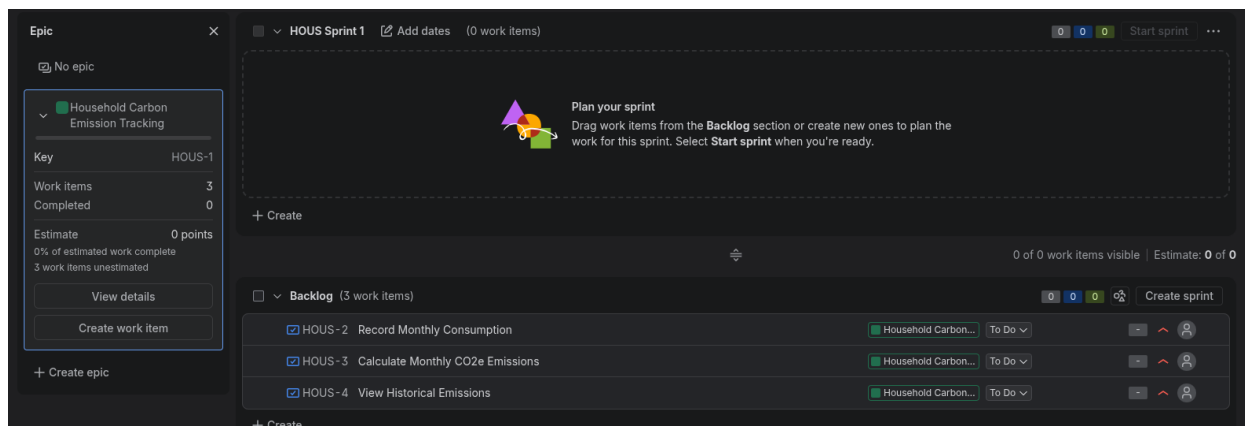


Fig 1.1 - Epic backlog with child User Stories and story points (sprint planning view)

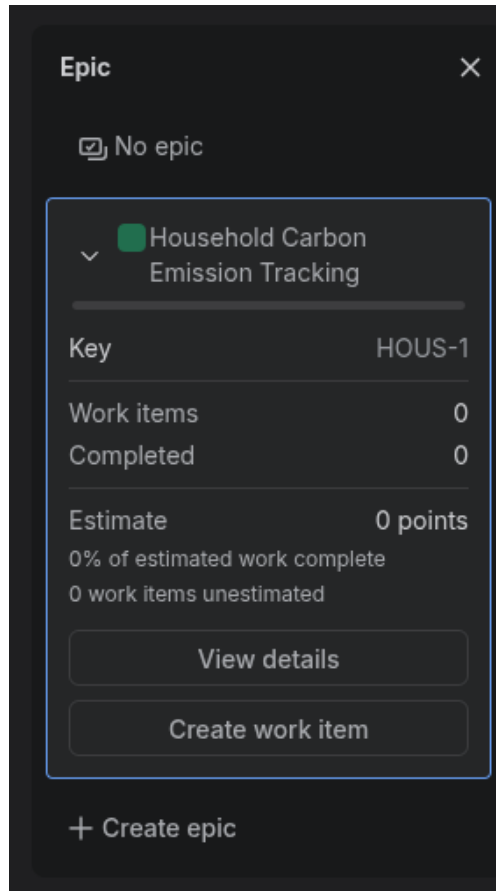


Fig 1.2 - Epic details panel (HOUS-1) showing associated backlog items

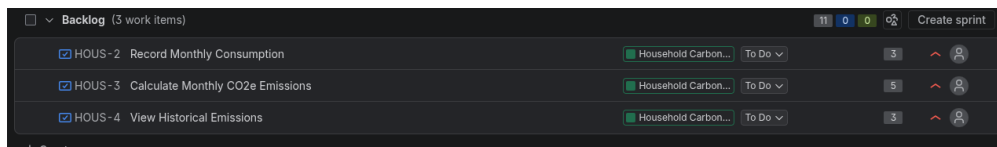


Fig 1.3 - Backlog view showing all three User Stories with story point estimates

Sprint 1 – Active Sprint Board

Sprint 1 included all three User Stories: Record Monthly Consumption (HOUS-2) was moved to Done, Calculate Monthly CO₂e Emissions (HOUS-3) was In Progress, and View Historical Emissions (HOUS-4) was in To Do.

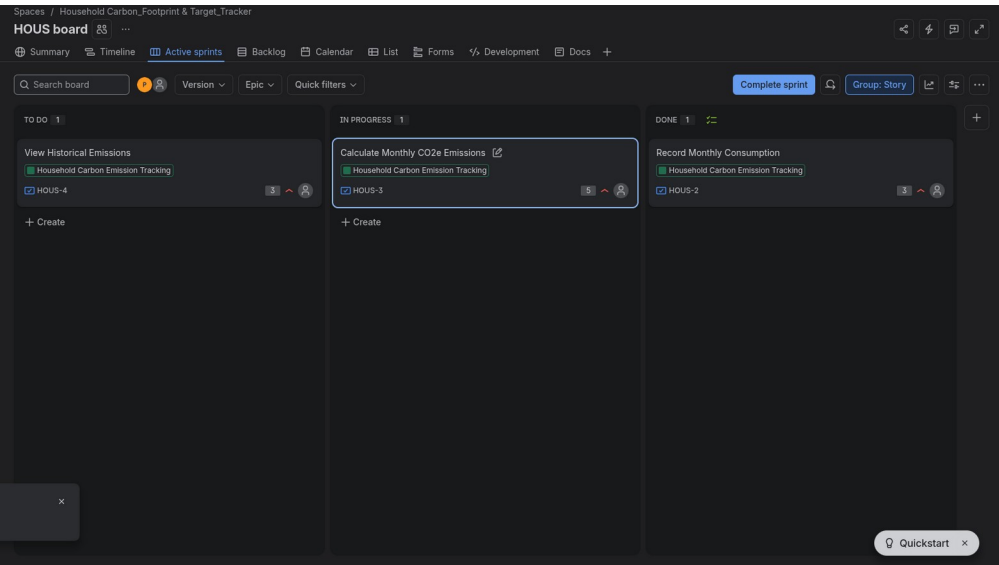


Fig 1.4 - Sprint 1 board (To Do / In Progress / Done)

Sprint 1 Burndown Chart

The burndown chart below shows the reduction in remaining story points as issues were completed during the sprint. All 11 story points were burned down within the sprint window, indicating the planned workload was achievable.

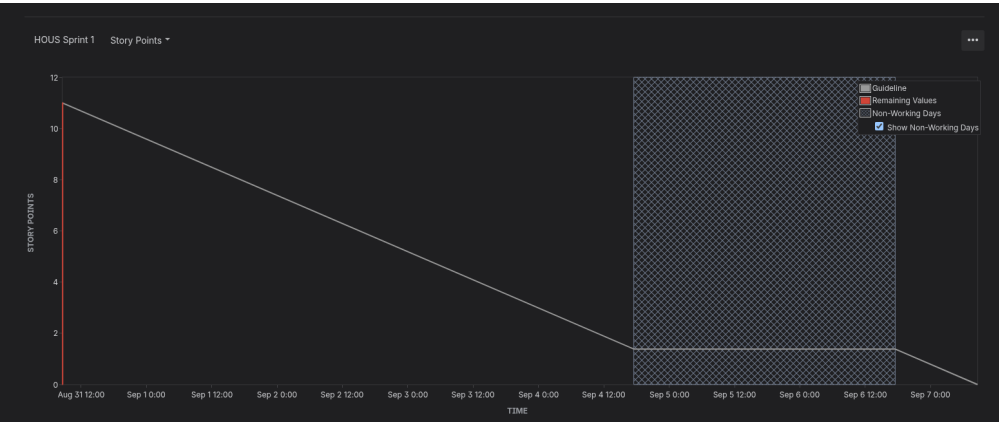


Fig 1.5 - Sprint 1 burndown chart (HOUS Sprint 1, Story Points)

Date	Issue	Event Type	Event Detail	Story Points		
				Inc.	Dec.	Remaining
31/Aug/26 8:36 AM	HOUS-2	Sprint start		3		
	HOUS-3			5		
	HOUS-4			3		
	Total					11
31/Aug/26 8:37 AM	HOUS-2	Burndown	Issue completed	3	8	
	HOUS-3		Issue completed	5	3	
	HOUS-4		Issue completed	3	0	
	HOUS-2	Sprint ended by prajwal			0	
	HOUS-3				0	
	HOUS-4				0	
	Total					0

Fig 1.6 - Sprint 1 burndown event log showing issue completion events

Reflection Questions

1. Did your estimations reflect the actual effort?

Yes. The estimated story points reasonably reflected the relative effort of the stories. The 5-point story (Calculate Monthly CO2e Emissions) involved more logic and complexity — applying emission factors to multiple consumption inputs and summing them into a CO2e figure — compared to the two 3-point stories (Record Monthly Consumption and View Historical Emissions), which were more straightforward data entry and read/display operations. The relative sizing ($5 > 3$) felt appropriate given the computational and validation work involved in the calculation story. In a real sprint, the 5-point story would be expected to take noticeably more time than either 3-pointer, and the estimates captured that correctly.

2. Was your backlog well-prioritized?

Yes. The stories were prioritized based on their dependency and user value. Recording consumption data (HOUS-2) was completed first because the calculation functionality (HOUS-3) directly depends on having that input data available. View Historical Emissions (HOUS-4) was correctly placed last, since it depends on both recording and calculating being functional before historical data exists to display. This dependency ordering meant the sprint naturally progressed through the stories in a logical sequence, delivering usable increments at each step rather than building dead-end features with no underlying data.

3. How did your simulated sprint align with your plan?

The sprint aligned closely with the plan. All three selected User Stories were completed within the one-week sprint window, resulting in all 11 story points being burned down by sprint close. The burndown event log confirms that issues were completed sequentially — HOUS-2 first, then HOUS-3, then HOUS-4 — which matched the intended priority order. The sprint ended cleanly with 0 remaining story points, indicating no carry-over into a subsequent sprint. In a real-team context, this would represent a well-scoped sprint where the team correctly estimated their capacity and delivered everything planned.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed a clear and steady reduction in remaining story points as each story was completed, reaching 0 by sprint end. Completing all 11 points within the sprint indicates that the planned workload was well-matched to the team's simulated capacity — neither over-committed nor under-committed. The chart's pattern (a relatively smooth decline ending at zero) reflects an ideal sprint outcome. In real projects, burndown charts are most valuable not just for tracking whether the team hits zero, but for catching irregular patterns early — such as mid-sprint spikes (indicating a story was prematurely marked Done and had to be reopened) or instant drops at sprint start (suggesting under-commitment during planning). The HOUS Sprint 1 chart did not exhibit these warning signs, which in a real context would suggest that the estimation and scope-setting process was reliable for this sprint.